University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid Term Examination, Spring 2025

Course Title: Data Structures and Algorithms I

Course Code: CSE0613211/CSE 211

Marks: 20 **Time: 1*(one)* hour**

(Answer all questions)

Marks

1. a) Briefly explain the difference between a singly, a doubly linked list, and a circular linked list. [03]

   b) Suppose you are designing a memory layout for a 2D array in row-major and column-major order. The base address of the array is 3900, and the array dimensions are 15x13, with each element being of type double (8 bytes). Compute the memory location of the element at location (14, 12) for both types of array representation. [03]
   Note- Array indexes must start from 1.

   c) Interpret the meaning of Big-O, Big-Ω (Omega), and Big-Θ (Theta) notations. For a given function $f(x) = 81x^3+98x^2+58$. Find the Big-Ω (Omega) for this function. [04]

2. a) Describe the four most frequently used data structure operations. [02]

   b) Write an algorithm to print out the following triangle. [03]

```
8
7 6
5 4 3
2 1 0 9
8 7 6 5 4
```

   c) Given an unsorted singly linked list: 2 → 10 → 3 → 8 → 6 → 5, construct an algorithm to sort the linked list in descending order using a singly linked list approach. [05]

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